

CSCE 775: Deep Reinforcement Learning

Homework 4

Your code must run in order to receive credit.

For extra credit, you will be using the DeepXube package to define a pathfinding domain, define a neural network that represents a heuristic function, train the heuristic function with machine learning, and use it to solve pathfinding problem instances. **Make sure to confirm the domain you are using with me as the domain cannot be something already implemented in the DeepXube framework.**

1 Visualization (10 pts)

Your domain should subclass `StateGoalVizable` and `StringToAct` so that problem instances can be visualized.

2 Test Problem Instances (10 pts)

You should have one or more .pkl files that contains problem instances for testing. These can be either randomly generated with DeepXube code or constructed manually. However, you must justify why they are interesting given your domain. For example, generating data with a random walk may only create problem instances that can easily be solved using uniform cost search. In such instances, more care needs to be taken when generating problem instances both for training and testing.

3 Train the Heuristic Function (15pts)

Train a heuristic function using any of DeepXube's methods for training.

4 Use the Trained Heuristic Function to Solve Problem Instances (15 points)

Use the trained heuristic function to solve problem instances using any of DeepXube's methods for solving. This should perform significantly better than uniform cost search.

5 Research (50pts)

More extra credit will be given for submissions that are addressing an unanswered research questions. For example, applying DeepXube to chemical synthesis, theorem proving, calculus, program synthesis, language models, image generation, etc.

What to Turn In

A zip file containing:

- A **single** Python file defining your domain and heuristic function.
- the .pkl files of the test problem instances.

- a README file describing how to train your heuristic function, how to use it to solve problems, and expected results.